

- 1** The photograph shows a brine shrimp that is found in salt lakes in North America.



(Source: © Nature Picture Library / Alamy Stock Photo)

These brine shrimps hatch from eggs and feed on algae.

Some salt lakes contain copper ions from industrial activity. The concentration of copper ions is increasing, as the lakes are drying out.

A student hypothesised that an increasing concentration of copper ions may affect the time taken for the brine shrimps to hatch from the eggs.

- (a) Describe an experiment to investigate the effect of the concentration of copper ions on the time taken for the brine shrimps to hatch from eggs.

(6)

(b) Suggest why using brine shrimps in this investigation does not create any ethical concerns.

(1)

(c) Copper ions (Cu^{2+}) can inhibit the enzymes involved in the hatching of brine shrimps.

Describe how copper **ions** could inhibit enzyme activity.

(3)

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(Total for Question 1 = 10 marks)